

gRPC MACH Architecture

TCO Costs Efficiency Calculator

Point-to-Point Integration Pattern · Protocol Buffer IDL · AWS Egress Pricing

✔ File 'unaryRPCPattern.proto' uploaded successfully!

A7 **Activity 7: Calculate the TCO Cost Considering Optimisation Strategies**

The Phase 4 report reflects the full optimised TCO: networking egress + cloud infrastructure – FinOps RI/SP savings. Unit economics, ARPU, ROI, break-even users, and the stakeholder PDF report are all generated here.



Phase 4 — TCO Networking Costs Report

Full cost breakdown including architecture tactics, unit economics, and stakeholder summary.

SERVICE UNDER ANALYSIS

order-m-s

pomoihuig

BUC1

1000 base RPS

350 End Users

–

APPLIED ARCHITECTURE TACTICS

Server-side LB Timeout Retry Circuit Breaker mTLS OAuth 2.0 + JWT

Applied Architecture Tactics & Patterns

INFORMATIONAL — NO RPS IMPACT

TACTIC	VALUE	DESCRIPTION	RPS IMPACT
Server-side Load Balancing	–	Balances the reception of gRPC packets across backend replicas.	None

TACTIC	VALUE	DESCRIPTION	RPS IMPACT
Timeout	0 ms	Discards lost packets and frees the client thread after the configured wait.	None
Circuit Breaker	Min calls: 0 Half-open calls: 0 Wait: 0 ms Failure threshold: 0%	Opens the circuit when failure thresholds are exceeded, protecting downstream services.	None
mTLS (mutual TLS)	0 reconnects/hr	Both client and server present X.509 certificates. CA validates both. 30: Max Payload overhead: bytes per message — RFC 8446 Because of RFC 8446 Section-5.2 * TLSCiphertext outer header: 5 bytes: * + 1-byte opaque_type=23 (uint8 ContentType) * + 2-byte legacy_record_version (uint16 ProtocolVersion) * + 2-byte length (uint16 length). * TLSInnerPlaintext prefix (AAD): * + 1-byte Content type and no padding. * AEAD Pre-record nonce: RFC 8446 Section-5.3 * + 8-byte length (64 bits explicit portion added during encryption). * AEAD tag: RFC 5116 Section-5.1 * + 16-byte tag appended to encrypted_record for AEAD_AES_128_GCM ciphertext is exactly 16 octets longer than its corresponding plaintext . Handshake is connection-scoped — negligible RPS impact at this reconnect rate. AWS ACM certificates are free.	None
OAuth 2.0 + JWT — bearer token	Unary interceptor TTL 3600s 1 client(s)	JWT Bearer header RFC 7515 Section-7.1 BASE64URL(UTF8(JWS Protected Header)) ' ' * BASE64URL(JWS Payload) ' ' * BASE64URL(JWS Signature) adds 650 bytes typical due to Payload overhead RFC 7519 §3 and RFC 9101: 650 bytes per message: Algorithm Standard claims RS256 Header 80 Payload (compact/std/ext) 200/400/600 Signature 344	None

TACTIC	VALUE	DESCRIPTION	RPS IMPACT
		Dots 2 Total 626/826/1026 → 650 typical . The token travels in every request's Authorization metadata header. Response messages carry no JWT overhead. Validation enforced by the configured gRPC interceptor.	

⚠️ RPS-IMPACTING — INCREASES NETWORKING COSTS

TACTIC	VALUE	DESCRIPTION	RPS IMPACT
Retry	2% max retries	Retries gRPC packet emission under temporary outages. Increases networking costs as up to N additional packets may be sent per logical request.	+20 req/s



RPS Adjusted by Active Tactics

BASE RPS	EFFECTIVE RPS
1,000	1,020



Security Byte Overhead Applied to Message Sizes

TLS OVERHEAD / MESSAGE	JWT OVERHEAD / REQUEST	EFFECTIVE REQUEST SIZE
30 bytes (typical)	650 bytes (typical)	938 bytes
30 / bytes typical	300 / 650 / 1200 bytes (min/typical/max)	
EFFECTIVE RESPONSE SIZE		
7577 bytes		

MESSAGE TYPE	PROTO SIZE (BYTES)	EFFECTIVE SIZE (BYTES)	REQUESTS / MONTH	THROUGHPUT (GB/MONTH)	TRANSFER COST (USD/MONTH)
Request	258	938	3B	2,309.6073 GB	—
Response	7547	7577	3B	18,656.6037 GB	\$ 1,637.01

🏠 Total Cost of Ownership (TCO) Breakdown

Cost Component	Category	Monthly Cost	% of TCO
AWS Egress (Response Transfer) 1000 eff. RPS · AWS data-out tiers	Networking	\$1637.01	87.6%
mTLS frame overhead (RFC 8446) Included in egress above — no additional charge	Security	\$0 (free)	free
JWT header overhead (RFC 7519 — request-side) AWS inbound data transfer is free	Security	\$0 (free)	free
Application Load Balancer (ALB) \$47.45/mo · /api/aws/alb-pricing	Cloud Infra	\$47.45	2.5%
Database / Backup / DR \$45.25/mo · /api/aws/database-backup-pricing	Cloud Infra	\$45.25	2.4%
Amazon Inspector \$3.53/mo · /api/aws/security-services	Security	\$3.53	0.2%
CloudWatch Logs \$53.00/mo · /api/aws/security-services	Security	\$53.00	2.8%
AWS Audit Manager \$6.00/mo · /api/aws/security-services	Security	\$6.00	0.3%
Containerized Cluster (EKS) \$73.00/mo	Cloud Infra	\$73.00	3.9%
API Gateway \$3.00/mo · /api/aws/api-gateway-pricing	Cloud Infra	\$3.00	0.2%
Total Cost of Ownership (TCO)		\$1868.24/mo	100.0%

* Costs computed via AWS Pricing API backend. Security costs: one row per selected service when volume entered. DR = Disaster Recovery | BC = Business Continuity.

 Unit Economics

TOTAL MONTHLY TCO (NETWORKING + CLOUD INFRA)

\$1868.24

\$1637.01 egress + \$231.23 cloud infra

 Per-Microservice Cost Metrics

TCO / MICROSERVICE /
MONTH

COST / REQUEST
\$0.00000100

ANNUAL PROJECTION
\$22418.88

\$1868.24

This service · full TCO

2.592.000.000 req/mo

12× monthly TCO

 Per-End-User Cost Metrics (350 End Users)

TCO / END USER / MONTH

\$5.3378

\$1868.24 ÷ 350 users

TCO / END USER / DAY

\$0.177927

Monthly ÷ 30

BUSINESS USE CASE

BUC1

Pattern cost attribution

 Revenue, ARPU & ROI Analysis

Revenue model: The value entered in Phase 1 is the *expected monthly revenue per end user* for this service. Total monthly revenue = that value × number of end users. **ARPU** = revenue per user per month. **ROI** = (total revenue – TCO) ÷ TCO. **Break-even users** = minimum end users needed to cover full TCO at current ARPU. Net Margin per user = ARPU – TCO per user. Present to Finance and Leadership to frame architecture cost as a revenue-generating investment.

TOTAL MONTHLY
REVENUE

\$2450

\$7.00/user × 350 users

ARPU / MONTH

\$7

Revenue per end user /
month

NET MARGIN / USER /
MO

\$1.6622

ARPU – TCO/user

MONTHLY ROI

31.1%

(Revenue – TCO) ÷ TCO

ANNUAL ROI

31.1%

12× monthly projection

BREAK-EVEN USERS

267

users to cover full TCO

REVENUE / \$1 INFRA

\$1.31

Revenue efficiency

ANNUAL REVENUE

\$29.400

12× monthly revenue

 FinOps Tagging & Labelling Strategy FINOPS BEST PRACTICE

 Tagging and labelling cloud resources is the foundation of cost visibility in FinOps. Without consistent tags, cost allocation across teams, services, and environments is impossible. The following strategy is derived from the **FinOps Foundation Tagging Best**

Practices and **AWS Cost Allocation Tags** guidance, applied to the cloud services selected for this microservice.

★ **MANDATORY TAGS (APPLY TO ALL SELECTED RESOURCES)**

Tag Key	Example Value	Purpose	FinOps Dimension
Environment	production staging dev	Separate cost by environment	Allocation
Team	platform-engineering backend	Charge back costs to the owning team	Chargeback
CostCentre	CC-1042	Finance cost centre mapping	Chargeback
Project	mach-ecommerce-replatform	Group costs by business initiative	Allocation
Application	order-management-service	Identify the owning application	Showback
ManagedBy	terraform cloudformation cdk	Track IaC ownership and drift	Governance
DataClassification	confidential internal public	Support security and compliance policy	Governance

🔧 **SERVICE-SPECIFIC TAGS (FOR THIS MICROSERVICE)**

Tag Key	Suggested Value	Purpose
ServiceName	order-m-s	Exact microservice identifier for cost drill-down
gRPCPattern	unary-rpc	The gRPC streaming pattern used by this service
BaseRPS	1000	Baseline load — supports capacity cost attribution
BusinessUseCase	BUC1	Ties cost to the business capability it supports
ProtocolBuffer	proto3	Serialisation format — relevant to egress cost model
FinOpsReviewed	2026-07-02	Date last reviewed by FinOps/architect for cost fitness

☁️ **SELECTED CLOUD SERVICES — TAGGING CHECKLIST**

☁️ **Security Services (GuardDuty / WAF / KMS / Macie / CloudWatch)**

- Application
- Environment
- Team
- CostCentre

 GuardDuty and Macie are account-level; use AWS Cost Categories for allocation. KMS CMKs: tag per owning service.

 **Governance note:** Tags are only effective when enforced via **AWS Tag Policies** (AWS Organizations) and monitored through **AWS Cost Explorer Tag Coverage** reports. Untagged resources should trigger automated alerts via **AWS Config Rules** (e.g. `required-tags`). Discuss tagging governance with your FinOps and Cloud Centre of Excellence (CCoE) team.

FinOps Architecture Notes & Disclaimers

1 — IP Address Planning (CIDR Sizing)

The more redundancy you build into AWS (Multi-AZ, replicas, NAT Gateways, ENIs), the more private IP addresses are consumed. Plan your VPC CIDR block generously (e.g. /16 = 65,536 IPs) and **leave enough address space within each subnet CIDR for auto-scaling headroom**. A /24 subnet provides only 251 usable IPs — exhausted quickly in a multi-tier, auto-scaled architecture.

2 — Multi-Tier Architecture in AWS Terms

Separating MACH components (frontend, API gateway, gRPC services, databases) into **different subnets** makes this a **multi-tier architecture** in AWS terms. Each tier needs dedicated subnets with its own Security Groups and NACLs.

3 — Intra-VPC Connectivity

Use **VPC Endpoints** (PrivateLink) to access AWS services without internet routing; **Transit Gateway** for cross-VPC connectivity; **VPC Flow Logs** for visibility. Each component in a separate subnet requires explicit route table entries.

* Outbound (response) data transfer cost uses AWS standard internet egress tiered pricing for US East (N. Virginia): first 10 TB at \$0.09/GB, next 40 TB at \$0.085/GB, next 100 TB at \$0.07/GB, beyond 150 TB at \$0.05/GB. Inbound (request) traffic is free. Effective sizes include TLS (RFC 8446) and JWT (RFC 7519) overhead where applicable. [AWS Pricing Calculator](#) ↗